

Examination of Blockchain-Based Content Monetization Models

This report furnishes a comprehensive examination of blockchain-based content monetization models, with an emphasis on NFTs and tokenization, as part of Weeks 3–4 of the thesis project plan. It draws upon literature and industry examples from 2025 to analyze mechanisms, real-world implementations, and gaps pertinent to the proposed cross-chain content rights management service on Polkadot. The analysis aligns with the research question by evaluating strategies for decentralized subscription, pay-per-view (PPV), and purchase models, emphasizing scalability, interoperability, and economic efficiency.

The review recognizes that blockchain monetization enhances creators' capabilities by facilitating direct peer-to-peer revenue, thereby reducing intermediary fees from 30–45% in centralized platforms to less than 5%. It also encompasses automated royalties via smart contracts and programmable scarcity. Market forecasts project the sector will attain a value of \$659 million by 2025, escalating to \$5.07 billion by 2032 at a compound annual growth rate of 33.61%, primarily driven by Web3 creator economies. Nevertheless, challenges such as insufficient cross-chain support and the absence of hybrid models, including subscriptions combined with pay-per-view (PPV), remain. The project seeks to address these issues through the implementation of ink! contracts and XCM.

1. Overview of Key Monetization Models

Blockchain models tokenize content as digital assets on immutable ledgers, using smart contracts for enforcement. Below is a breakdown:

Table 1: Key Blockchain Monetization Models

Model	Description	Key Mechanisms	Applicability to Project Models	Pros	Cons
NFTs	Unique tokens representing ownership of digital content (e.g., art, videos).	ERC-721/ERC-1155 standards; royalties (5–10% on resales).	Purchase: Full ownership transfer; PPV: Time-bound access NFTs.	Proven scarcity; perpetual royalties.	High minting fees; volatility.
Tokenization	Fractionalizing content into fungible tokens (e.g., ERC-20) for shared ownership.	Fractional NFTs; staking for yields.	Subscription: Recurring token burns; Purchase: Fractional shares.	Democratizes access; liquidity.	Complex governance; dilution risk.
Social Tokens	Creator-issued tokens for fan engagement and access tiers.	Utility tokens; governance voting.	Subscription: Tiered access; PPV: Token-gated views.	Community-driven revenue.	Regulatory scrutiny (securities).
Micro-Payments	Small, instant transactions for granular access.	Layer-2 solutions (e.g., Lightning).	PPV: Per-view fees (<\$0.01).	Low barriers; high volume.	Scalability bottlenecks.
Staking/Rewards	Lock tokens for yields or exclusive content.	DeFi yields; proof-of-contribution.	Subscription: Staked access renewals.	Passive income; loyalty.	Impermanent loss; lock-up periods.

2. Deep Dive into NFTs and Tokenization

NFTs in 2025:

- **Evolution:** Non-fungible tokens (NFTs) have transitioned from being mere collectibles to functional utilities, mediating access, fostering community engagement, and generating revenue within gaming and metaverse ecosystems. By 2025, multi-chain NFTs will facilitate seamless trading across Ethereum, Solana, and Polkadot, while the integration of fiat and cryptocurrencies is projected to increase adoption rates by 40%. Creators will derive earnings through primary sales (averaging between \$500 and \$5), royalties

(ranging from 8% to 10%), and secondary market transactions.

- **Monetization Mechanics:**

- **Purchase Model:** ERC-721 tokens grant perpetual ownership; smart contracts enforce resale royalties.
- **PPV:** Dynamic NFTs unlock time-limited views (e.g., 24-hour access).
- **Subscription:** NFT “memberships” auto-renew via oracles.

- **Examples:**

- **Gaming Creator Economy:** Platforms like Immutable X use NFTs for in-game assets, generating \$200M+ in 2025 royalties. Creators retain 95% via blockchain.
- **Metaverse:** Decentraland NFTs monetize virtual land/content, with \$150M market in Q3 2025.
- **Cross-Chain:** Axelar-powered marketplaces allow NFT transfers between Polkadot parachains and Ethereum, reducing fees by 70%.

Tokenization in 2025:

- **Evolution:** Fractional tokens enable micro-ownership (e.g., 1/100th of a video), with \$1.2B tokenized content market. Social tokens (e.g., Rally.io) blend utility and governance.

- **Monetization Mechanics:**

- **Subscription:** Tokens as “keys” burned monthly for access.
- **Purchase:** Fractional NFTs for high-value content (e.g., films).
- **PPV:** Pay-per-token for segments.

- **Examples:**

- **Fan Engagement:** Creators on Mirror.xyz tokenize newsletters; \$50M earned in 2025 subscriptions.
- **Music:** Audius tokenizes tracks for fractional royalties, with cross-chain via Cosmos IBC.

Cross-Chain Implementations:

- **Examples:**

- **Fourchain NFT Marketplace:** Supports Polkadot-Ethereum bridges for content NFTs; handles 10K+ daily trades.
- **Calibraint Cross-Chain Architecture:** Uses XCM-like protocols for automated swaps; 2025 case: Video rights transfer in 2s.
- **Oodles Cross-Chain Marketplace:** Integrates 5+ chains for PPV NFTs; 60% fee reduction.

3. Gap Analysis

Despite growth, 2025 models reveal critical gaps, identified via literature synthesis:

- **Interoperability Silos:** Seventy percent of NFTs are locked on Ethereum; cross-chain transfers encounter failures in 15–20% of cases due to vulnerabilities in bridges. This presents an opportunity for the project: Native XCM addresses this issue by enabling seamless subscriptions across Polkadot and Ethereum.
- **Subscription and Pay-Per-View Limitations:** Only a quarter of platforms support recurring revenue models; unclear billing practices diminish consumer trust (for instance, hidden charges are present in 40% of Web3 subscriptions). Pay-Per-View micro-transactions encounter scalability constraints, with transaction speeds limited to 15 transactions per second on the Ethereum network. The project addresses these issues: ink! contracts facilitate automated renewal processes with latency of less than two seconds.
- **Economic Inefficiencies:** Elevated transaction fees (\$0.05–0.50 per transaction) discourage pay-per-view models; fractional token dilution diminishes creator revenues by 20%. Regulatory ambiguities, such as SEC scrutiny of social tokens, obstruct widespread adoption.
- **User Experience:** Complex wallets fragment engagement; 60% abandonment in cross-chain flows. Gap: Centralized UX hybrids needed.
- **Scalability and Security:** Single-chain models are limited to 100 transactions per second; the 2025 hacks are estimated to have incurred costs of \$500 million. The project's advantage lies in Polkadot's capability to process over 1,000 transactions per second and its shared security model.
- **Adoption Barriers:** The mass market lacks fiat on-ramps; only 15% of creators utilize blockchain technology due to gaps in expertise.

Quantitative Gaps Table (Based on 2025 Benchmarks):

Table 2: Quantitative Gaps (2025 Benchmarks)		
Gap Area	Current Metric (2025 Avg.)	Project Target Improvement
Cross-Chain Success Rate	80%	99% (XCM)
PPV Tx Cost	\$0.10	\$0.02
Subscription Churn	35% (opaque billing)	<10% (transparent on-chain)
TPS for High Load	100 (Ethereum)	500+ (Parachains)

These gaps confirm the project's novelty: No 2025 framework fully integrates cross-chain subscriptions/PPV/purchases with Polkadot efficiency.

4. Relevance to the Thesis Project

- **Informs Design:** NFTs/tokenization map to PurchaseVerification.rs (ERC-721-like) and SubscriptionManager.rs (token burns).

- **Evaluation Metrics:** Benchmarks for economic efficiency (60% savings vs. centralized) and interoperability (2s latency).
- **Fills Gaps:** XCM enables multi-chain PPV; shared security reduces hacks; Rust/ink! ensures low fees.
- **Contributions:** Novel hybrid model boosts creator retention to 95%, addressing 2025's \$12B piracy losses.

Implementation Notes (Phase 4–5 Measured Results)

The targets and claims above were Phase 1 theoretical projections. Phase 5 testing produced the following measured results:

Table 3: Phase 1 Targets vs Phase 5 Measured Results

Phase 1 Target	Measured Result	Assessment
TPS: 500+ (Parachains)	27 TPS sustained; 283 TPS theoretical per parachain	Single-parachain throughput is bounded by relay chain slot allocation. Polkadot scales horizontally via more parachains.
Cross-Chain Success Rate: 99%	100% success (9 XCM operations, all succeeded)	Exceeded target — though measured sample was smaller than planned
PPV Tx Cost: \$0.02	Transaction weight fees are minimal (~0.000001 DOT equivalent)	Achieved — on-chain fees are negligible
Latency: <2 seconds	~6 seconds (single-chain), 18–32 seconds (cross-chain XCM)	Not achieved — 6s block time is inherent to the parachain model
Economic efficiency: 60% savings	Self-publishing model eliminates intermediary entirely (0% commission vs 15–30% traditional)	Exceeded — but via a different mechanism (disintermediation, not Monte Carlo optimisation)
Architecture: ink! contracts	FRAME pallet + ink! API layer	Pivoted — FRAME pallet provides better performance and direct storage access
NFT standard: RMRK 2.0	pallet-nfts with attribute-based nesting	RMRK 2.0 was abandoned by the community

Bibliography

1. Blockchain Content Monetization Models: How Creators Earn in Web3. (2025). *Indspn.org*. <https://indspn.org/blockchain-content-monetization-models-how-creators-earn-in-web3>

Annotation: Provides an overview of NFTs and social tokens; \$659M market forecast for 2025. Important for subscription benchmarks; identifies gaps in PPV scalability.

2. NFTs as Token Utilities in 2025. (2025). *DigitalCurrencyTraders*. <https://digitalcurrencytraders.com/nfts-as-token-utilities-in-2025-rethinking-community-access-and-revenue-cbb937158e7b>

Annotation: NFTs as a means of access; there has been a 40% increase in adoption. It supports Pay-Per-View (PPV) and highlights deficiencies in user experience (UX).

3. How Gaming and Blockchain Are Powering the Creator Economy in 2025. (2025). *EdgeofNFT*. <https://www.edgeofnft.com/podcasts/how-gaming-and-blockchain-are-powering-the-creator-economy-in-2025>

Annotation: Key Takeaways

- Blockchain and gaming are creating new revenue streams for creators in 2025.
- NFTs ensure ongoing royalties and authentic asset ownership in games.
- Web3 gaming supports creators via DAOs, play-to-earn, and token rewards.
- Revenue ideas include in-game crypto economies, community incentives, and interoperable assets.
- NFT innovations are transforming creator empowerment.

4. 10 Metaverse and NFT Trends 2025–2026. (2025). *Beinsure*. <https://beinsure.com/metaverse-nfts-trends-outlook/>

Annotation: Analyzing trends in the digital domain and the metaverse provides a strategic advantage, as numerous technology corporations are investing in blockchain technology, emphasizing its future significance. Non-fungible tokens (NFTs) possess considerable potential within the metaverse for purposes such as ownership, scarcity, and monetization. Challenges associated with representing ownership and managing inflation are present; however, emerging solutions are being developed. Overcoming these obstacles could position NFTs as a fundamental component of virtual environments.

5. Advanced Strategies for NFT Marketplace Architecture: Scalability, Security & Cross-Chain Innovation. (2025). *Calibraint*. <https://www.calibraint.com/blog/advanced-nft-marketplace-architecture>

Annotation: NFTs have shifted from collectibles to a booming digital economy encompassing gaming, real estate, fashion, and music. Statista reports the global NFT market may hit \$2.37 billion in 2024, growing 9.10% annually through 2028. Businesses entering the space often ask: how to design scalable platforms, secure user data, and support cross-chain trades? This blog explores key strategies and technical principles to future-proof NFT marketplace architecture.

6. Cross-Chain NFT Marketplace Development: The Future of Digital Ownership. (2025). *Fourchain*. <https://www.fourchain.com/nfts/cross-chain-nft-marketplace-development>

Annotation: NFTs have transformed perceptions of digital ownership. The development of cross-chain NFT marketplaces enhances this transformation by allowing cre-

ators, artists, and collectors to monetize and trade digital assets across various blockchain networks. The expanding NFT ecosystem encounters the challenge of assets being distributed across multiple blockchains, with cross-chain marketplaces providing seamless transaction capabilities. This guide encompasses development processes, technical considerations, benefits, use cases, and practical applications of these marketplaces.

7. What Is a Cross-Chain NFT Marketplace And Why It Matters in 2025. (2023/2025 Update). *Oodles*. <https://blockchain.oodles.io/blog/cross-chain-nft-marketplace/>

Annotation: Cross-chain NFT marketplaces are the next evolution in Web3. They allow users to mint, trade, and move NFTs across multiple blockchains seamlessly. This guide explores how they work, the tech powering them (bridges, oracles, smart contracts), key examples, and the real-world impact.

8. Crypto Payments for Content and the Future of Broadcasting Monetisation. (2025). *ResearchGate*. https://www.researchgate.net/publication/396401461_Crypto_Payments_for_Content_and_the_Future_of_Broadcasting_Monetisation

Annotation: This paper explored cryptocurrency payments and blockchain as innovative tools for broadcasting monetization. Using academic sources, industry reports, and case studies, it identified opportunities like micropayments, token ecosystems, transparent royalties, and global access. The findings show cryptocurrency can democratize content and enhance creator-audience relationships through low-cost, borderless transactions. Challenges include volatility, regulation, scalability, and user adoption. The study concludes that, while cryptocurrencies likely won't replace traditional models soon, hybrid systems with fiat and crypto payments are a practical solution.

9. Subscription vs. Pay-Per-View: Best Revenue Model. (2025). *Oyelabs*. <https://oyelabs.com/subscription-vs-pay-per-view-revenue-model-for-content-creation-app/>

Annotation: SVOD services like Netflix and Hulu offer unlimited content for a monthly fee, while PPV is ideal for live sports and events. Each has strengths and limitations, and not all content fits every model. This blog explores how they work, their benefits, and challenges, helping you select the best approach for your content and audience to build a sustainable platform.

10. Blockchain for Content Monetization Market Size 2025–2032. (2025). *360iResearch*. <https://www.360iresearch.com/library/intelligence/blockchain-for-content-monetization>

Annotation: This executive summary examines the drivers of blockchain adoption in content monetization, industry shifts, and how external factors, such as U.S. tariffs, affect costs. It also analyzes segmentation across content types, architectures, models, applications, end users, regions, and competitors.

11. Blockchain-Based Subscription Services. (2025). *Medium*. <https://medium.com/%40genxaiblogs/blockchain-based-subscription-services-genx-ai-paving-the-way-for-a-new-era-of-recurring-value-cd23ab6d8fc6>

Annotation: GenX AI: Rethinking Subscription Models in the Digital Age. Subscription services dominate industries like streaming, SaaS, boxes, and digital publications, but face challenges such as opaque billing, high fees, limited control, and centralization, hindering innovation. Blockchain provides decentralized, secure, transparent solutions to transform subscriptions. With GenX AI, blockchain platforms aim to make subscriptions more flexible, fair, and user-centric.

12. NFTs in 2025: A Market Reborn. (2025). *Medium*. <https://medium.com/predict/nfts-in-2025-a-market-reborn-or-a-forgotten-trend-04led7a8234c>

Annotation: The tech world evolves rapidly. Yesterday's breakthrough can quickly be overshadowed, as NFTs were a 2020s buzzword for celebrities, artists, and investors. After the hype burst, many doubted NFTs, thinking they were just a trend. But by 2025, NFTs will have evolved beyond digital art, influencing industries and redefining ownership. They're proving to be more than fleeting fads.

13. How Blockchain Is Revolutionizing Fan Engagement. (2025). *iLink*. <https://ilink.dev/blog/how-blockchain-is-revolutionizing-fan-engagement-and-monetization-for-creators/>

Annotation: In today's digital age, creators face new challenges and opportunities in engaging audiences and monetizing their work. Traditional methods like merchandise, ticketing, and streaming have served well but have limitations, including reliance on intermediaries, opaque revenue-sharing models, and limited control over fan interactions. Blockchain offers decentralized ways for creators to connect directly with fans and unlock new monetization avenues. This article explores how blockchain is transforming fan engagement and monetization.

14. Blockchain Creator Economy: 7 Trends 2025. (2025). *Avanti3*. <https://avanti3.com/blockchain-creator-economy/>

Annotation: The blockchain creator economy revolutionizes how creators monetize and engage with fans using decentralized tech. It replaces platform cutthroat models with direct creator-fan relationships, enabling transparent payments and content ownership. Features include minimal fees (1–2.5% vs. 30–50%), smart contracts for revenue sharing, NFTs for royalties, and fan participation via tokenization and governance.

15. Monetization For Blockchain Projects. (2025). *Meegle*. https://www.meegle.com/en_us/topics/monetization-models/monetization-for-blockchain-projects

Annotation: Blockchain has revolutionized industries with decentralized, transparent, and secure solutions. However, creating sustainable revenue streams remains a challenge for many projects. Monetization isn't just about income; it's about long-term viability and delivering value. Whether you're an entrepreneur, developer, or investor, understanding monetization is vital. This guide explores strategies, challenges, and opportunities, providing insights and examples to navigate this complex landscape.

16. The Future of Content Monetization: How Emerging Payment Solutions Are Reshaping the Creator Economy for Investors. (2025). *AInvest*. <https://www.ainvest.com/news/future-content-monetization-emerging-payment-solutions>

ions-reshaping-creator-economy-investors-2509/

Annotation: Emerging payment solutions, such as micropayments and blockchain, are reshaping the creator economy by enabling real-time monetization and global access. Platforms such as Coil and Brave leverage blockchain for frictionless micropayments, while NFTs on Zora and Base Chain redefine ownership. North America and Asia-Pacific lead adoption, with blockchain payments expected to grow 12.6% CAGR, driven by stablecoins and mobile wallets. Investors face challenges such as regulatory uncertainty but find opportunities in platforms that blend AI fraud prevention with scalable, compliant infrastructure.

17. NFT Marketing Trends 2025. (2025). *BlockchainAppFactory*. <https://www.blockchainappfactory.com/blog/nft-marketing-trends-2025/>

Annotation: Non-Fungible Tokens (NFTs) have risen from curiosity to a key part of the digital economy. They offer a unique, verified way to own and trade digital assets, making them valuable for marketing. Businesses are quickly recognizing their potential.

18. Ultimate Guide to ERC-721 2025. (2025). *RapidInnovation*. <https://www.rapidinnovation.io/post/what-is-erc-721>

Annotation: Standards; Polkadot compat. For ink! mapping.

19. 27 Top Blockchain Applications 2025. (2025). *G2*. <https://learn.g2.com/blockchain-applications>

Annotation: When you hear “blockchain,” you might think of Bitcoin or cryptocurrencies, but it offers much more. At its core, blockchain records, shares, and protects information without relying on a single intermediary, impacting various businesses. If you’re curious about how companies use blockchain beyond crypto headlines, this guide introduces notable real-world applications—from tracing products in supply chains to smart contracts and healthcare data security—showing how blockchain is transforming trust and value creation.

20. How Cross-Chain Technology is Shaping the Future of Digital Business. (2025). *SecuritySenses*. <https://securitysenses.com/posts/how-cross-chain-technology-shaping-future-digital-business>

Annotation: Before cross-chain tech, blockchains operated independently, restricting data, tokens, and assets to their native systems, limiting interoperability, scalability, and usability. As the ecosystem grew, these limits became clear. Cross-chain tech was developed to overcome this by enabling seamless exchanges between different blockchains. To understand their potential and impact on the digital landscape, all aspects of this article warrant exploration.

21. NFT Intellectual Property Rights. (2025). *Meegle*. https://www.meegle.com/en_us/topics/web3/nft-intellectual-property-rights

Annotation: NFTs (Non-Fungible Tokens) transform digital ownership in Web3. An NFT is a unique digital asset verified via blockchain, confirming scarcity and ownership. This impacts intellectual property laws and opens new opportunities for creators, developers, and businesses. Digital artists and musicians use NFTs for ownership and authenticity, enabling limited editions with proof of ownership, and providing direct in-

come and control over use and resale. For businesses, NFTs redefine product ownership and customer engagement; in gaming, they let players own tradable in-game items with real-world value. Applications include virtual real estate and digital fashion. NFTs challenge traditional copyright laws, making it essential for professionals to understand smart contracts that govern ownership. Overall, NFT rights increase opportunities for creators and firms, fostering innovation and new economic models, while highlighting the importance of tech and legal understanding.

22. Revolutionizing Content Creation and Monetization with Web3. (2024/2025). *Block-Apps*. <https://blockapps.net/blog/revolutionizing-content-creation-and-monetization-with-web3/>

Annotation: The content creation world is on the verge of a transformation driven by Web3 technologies, marking a shift towards decentralization, user ownership, and community-driven platforms. Content creators must understand Web3 concepts to navigate and benefit from this evolving landscape.

23. Scarcity-Driven Monetization. (2022/2025 Update). *PMC*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9441551/>

Annotation: Over the past two decades, the U.S. news industry faced a 66% revenue drop, prompting publishers to try new revenue strategies. Some, like paywalls and freemium models, boost income but limit news access, making news scarcer. Others, like reader-focused fundraising, aim to increase revenue organically without reducing news availability. This chapter examines these digital monetization approaches, evaluating the sustainability of scarcity-driven strategies, conditions for profitability, and future industry trends.